● ahsanul —full

# Ahmad Abdul-Jawad Mahmood

**Full-Stack Engineer** | Founder & CTO, Buraq

// backend · web · mobile · 12+ years in production · Cairo · Riyadh · database to App Store

| 9 | 330+ | 2,900+ | 12+ |
|---|---|---|---|
| countries · BY ONE AUTHOR | companies | users · TESTS | years · ENGINEERING |

```
/**
 * Full-Stack Engineer — I own a feature from the database to the
 * app on the phone. Since March 2020: Founder & CTO of Buraq (buraqsupport.com),
 * an Egyptian ride-hailing platform to build solo across every layer — Next.JS /
 * TypeScript backend on PostgreSQL and Redis (22 modules, 330+ endpoints,
 * p.99ms rest), a Jetpack Compose driver app with maps, chat, tracking, native
 * Socket.IO feeds, payments, and native iOS & Android apps on one API contract.
 * Sorted it: 3 years of production engineering, seven of them running a
 * POS used by 200+ restaurants.
 */
```

## across_the_stack()

4 pillars

**[01] backend & web**

### Full-stack TypeScript

Next.JS 14 · Express · tRPC · Zod · Prisma · PostgreSQL · Redis · Socket.IO · Tailwind · shadcn/ui

Owned the whole stack: 22 modules on an exotic graph — trip state machine, scored waterfall dispatch over a Redis geo index, admin console pricing across six time tiers, double-entry ledger with cash and card settlement (Paymob, Fawry), Socket.IO rooms for live tracking, and Expo push. 330+ endpoints, p.99 of 7ms rest / 4ms tRPC, Sonarqube and audit log. 330+ endpoints over 55 controllers, 42 entities, 97% unit tests. Swagger docs.

**[02] mobile**

### Native Android & iOS

Jetpack Compose · Material 3 · Hilt · Retrofit · Socket.IO · Google Maps · Firebase · Swift · SwiftUI · Combine · MapKit · Keychain

Shipped both clients in one week: Android / Compose apps on shared, versioned core libraries, with 3,500+ LOC and 4,000+ Android tests. Before this: apps used by 2,900+ riders & drivers. Navigation, Epo ride-hailing and the Nuru POS.

- Android / Compose — 3,500+ client tests
- SwiftUI / Combine — 4,000+ client tests

**[03] web**

### Ops console & customer site

Next.JS · React · Tailwind · shadcn/ui · Recharts · Leaflet · Vercel

**12 years of shipping apps**

Ops console for live dispatch, pricing, and support. Customer site for booking, tracking, and ratings. Deployed on Vercel.

- Ops Console — dispatch, pricing, support
- Customer Site — booking, tracking, ratings

**[04] web**

### An ops console people can run a business on

Next.JS · React · Tailwind · shadcn/ui · Recharts · Leaflet · Vercel

A 29-screen Next.JS 14 console: live Mapbox trips map with demand heatmap, KYC review, finance reconciliation, refunds and payouts, a map zone editor, feature flags, RBAC and an audit log hashing Google SSO.

- 29 screens · RBAC · audit log

**[05] web mobile clients**

### Three products that talk to the world

Socket.IO gateways for riders, drivers and admins; Google Maps routing with ETA and geofencing; Paymob and Fawry payments; Twilio and Firebase messaging. Whatsapp-first OTP with SMS fallback. RTMs and FCM push.

- Sockets · maps · payments · OTP · push

## experience[]

9 roles

### Founder & CTO — Backend · Web · iOS · Android @ Buraq — buraqsupport.com

Mar 2024 — Present · CURRENT

CAIRO · FOUNDER · BURAQ TECHNOLOGIES

Founded an Egyptian ride-hailing platform and have carried it **entire technical scope alone**: chose the stack, set the architecture, and built nine repositories solo — a hardened NestJS backend and operations console, rider and captain apps on **iOS and Android** in Jetpack Compose, and a marketing site at **buraqsupport.com**. Every layer governed by explicit rules: idempotent mutations, double-entry accounting, explicit module boundaries, and a CI suite that gates the merge. Now preparing a cash-only Loan beta in Cairo.

- Modular monolith, **22 modules** on an exotic graph — **trip state machine**, **scored waterfall dispatch** over a Redis geo index, admin console pricing across six time tiers, **double-entry ledger** with cash and card settlement (Paymob, Fawry), Socket.IO rooms for live tracking, and Expo push. **330+ endpoints** over 55 controllers, 42 entities, **97% unit tests**. Swagger docs.
- **29-screen** ops surface — live **Mapbox** trips map with demand heatmap, KYC review, finance reconciliation, refunds and payouts, a map zone editor, feature flags, RBAC and an audit log hashing Google SSO. English and Arabic. 370+ tests.
- Two native **SwiftUI** apps on a shared **BuraqCore** package — networking, persistence, Keychain, design system, network-aware **offline queue** with **CRDT**-style conflict resolution, and background location, silent push, and payments. **Live Activities**. Siri Voice Shortcuts — one-tap "Ride to Work", captain earnings and navigation · 240 views, **1,800+ tests**, strict concurrency.
- Native apps **powered** the **same API contract** — 3 Gradle modules and 3 packages, 3,500+ LOC, zero duplicated domain; onboarding, the eKYC 9-Persona service and the trip and by end running against the backend · **3,500+ tests**.
- Screen-by-screen parity port of the iOS rider app — 11 feature modules on a published **buraq-core-android** library. **Material 3** theming, RTL-first, 4,000+ tests in English, Arabic, and RTL.
- Marketing site, **rider web app** with live trip tracking, **captain web app** with on-demand document KYC, public trip-share pages and sign-up flows · English and Arabic (RTL). MIT. 2 2 2K · **1,800+ tests** · **live on Render since 15 Oct 2026**.

One backend · one contract · six client surfaces · and the company around them. Incorporation, the Egyptian tax and labor regime, the Google Play and App Store records, the business model and investor materials, the go-live plan. 3,100+ commits, one author — and I'd have another shot at every one of the hard ones.

### Team Lead @ Marn — Jahez Group

Aug 2019 — Aug 2020

RIYADH

- **Led a 5+ engineer iOS team** for seven years — onboarding, mentoring, career growth, and the technical bets behind an enterprise POS used by **200+ restaurants** and cafés.
- **Owned the technical roadmap** and architecture: an offline-first CoreData core with cloud sync, built on a worker keeps trading when the network does not.
- **Set the engineering bar**: module boundaries, code review, and a unit-testable testing — holding the app at a **99.8-99.9% crash-free** rate in production.
- **Drove the multi-year migration** to SwiftUI, shipping it without forcing feature delivery, and made the case for it to product leadership.
- **Built CI/CD and the release train from scratch**, cutting deployment time **-63%** and shortening every engineer's feedback loop.
- Ran the performance practice (**cold start load times**) and designed the table-management, order-processing and inventory systems at the product's core.
- Partnered with product, design and backend leadership on scope, estimates and delivery commitments.

### Senior iOS Engineer @ HarvardEur · voluntary charity project

Dec 2025 — Jun 2026

- Designed an architecture integrating **on-device machine learning** for real-time content analysis.
- Built a seamless communication layer between native iOS and web-based content.
- Implementing offline-first data synchronization to ensure a consistent without causing feature delivery, and made the case for it to product leadership.
- Partnered with product, design and backend leadership on scope, estimates and delivery commitments.

### iOS Team Lead @ Epo — Ride Hailing

Mar 2018 — May 2019

RIYADH

- Developed a ride-hailing ecosystem (iOS, watchOS) serving **10M+ users** in Saudi Arabia.
- Implemented real-time GPS tracking and navigation with Mapbox and optimized route algorithms.
- Integrated Apple Pay, HyperPay, and push-notification infrastructure.
- Created seamless iPhone → Apple Watch handoff for continuous ride tracking.
- Built the driver rating system and real-time fare calculation engine.

### Senior iOS Engineer @ RungeStation — DeliveryHero

Sep 2016 — Mar 2018

BERLIN

- Contributed to a major Food-delivery platform serving **20M+ users**.
- Optimized app performance for high-volume transaction processing.
- Implemented real-time order tracking and delivery status updates.
- Collaborated with backend teams on API design and integration.

### iOS Developer @ ALAqeeh

May 2016 — Aug 2016

RIYADH

- Developed food-ordering apps with Firebase integration and real-time features.
- Implemented Stripe payments and live-chat functionality.
- Built responsive UI with Auto Layout across multiple device sizes.

### iOS & Apple Watch Developer @ Nite Design Co.

Jul 2015 — Sep 2015

- Early adopter of Apple Watch development (watchOS 1-2).
- Built a foundation in iOS and watchOS app development.
- Adopted SwiftUI from BNDC 2019 onward.

### iOS Developer @ UK Pro. Solutions Ltd.

May 2014 — May 2015

- Founded the iOS experience in Objective-C and early Swift.
- Built social networks and e-commerce platforms.
- Learned core iOS principles and App Store deployment.

## projects[]

selected

**[01] iot platform**

### HarvardEur — Privacy Browser

SwiftUI · CoreML · WKWebView · CoreImage · Vision

Built a seamless communication layer between native iOS and web-based content. Implemented offline-first data synchronization to ensure a consistent without causing feature delivery, and made the case for it to product leadership.

- Created a real-time video monitoring system for continuous content filtering.
- Developed smart banner blocking and optimized memory management throughout.

**[02] fintech**

### RungeStation

ECS25 · Firebase · Real-time APIs · Perf

Optimized app performance for high-volume transaction processing. Collaborated with backend teams on API design and integration.

- Contributed to a major Food-delivery platform serving 20M+ users across Saudi Arabia and Bahrain.
- Implemented real-time order tracking and delivery status updates.

★ 4.5 / 5 — 1.4M+ ratings

**[03] enterprise**

### Marn — iPad POS

SwiftUI · CoreData · CloudKit · Real-time Sync

Enterprise POS serving 200+ restaurants across Saudi Arabia and Bahrain. Offline-first architecture, complex table management, real-time orders.

- Built the driver rating system and real-time fare calculation engine.

★ 4.8 / 5 — 2,417 ratings

**[04] marketplace**

### Nuru — Ride-Hailing

RideSDK · TypeScript · PostgreSQL · Redis · Docker CE · Next.JS · SwiftUI · Kotlin · Jetpack Compose

Egyptian ride-hailing platform I founded and built solo: a hardened NestJS backend (330+ endpoints, scored dispatch, double-entry ledger, live tracking), native rider and captain apps on iOS and Android, and the public web app live at buraqsupport.com.

- Founder & CTO · nine repos · 10k+ tests

**[05] fintech**

### Epo — Ride Hailing

MapKit · Core Location · Social SDKs

Ride-sharing platform with Google Maps integration and social-media login.

- Built the driver rating system and real-time fare calculation engine.

★ 4.5 / 5 — 211.8m ratings

**[06] consumer**

### Dalaala

Dalaala · Core Location · Social SDKs

Mumlia community app — grower times, harvest times, and community features with enhanced location + social integration.

★ 4.5 / 5 — 1,187 ratings

## stack.config

```
web_and_infrastructure =
  Next.JS 14 · React · TypeScript · MapLibre
  Socket.IO · Docker · GitHub Actions CI
  Prometheus Metrics · Jest · Google Maps Platform
  Paymob · Cloudflare R2

ios_2025 =
  SwiftU · A D · SwiftTest · ECS25 · Combine · AsyncAwait
  SwiftData · ECS 26+
  Socket.IO · Swagger

android_2025 =
  Battle · Jetpack Compose · Android · Coroutines
  Flow · MVVM

architecture =
  MVVM · ECMAR · Clean Architecture · Coordinator
  TDD

apple_ecosystem =
  watchOS · tvOS · macOS · Widgets
  Live Activities · App Intents

intelligence =
  Core ML · Vision · ARKit · Dynamic Island
  App Tracking

backend_mcls =
  Next.JS · TypeScript · PostgreSQL · TypeORM · Redis
  JWT/tPP · REST · GraphQL · Firebase · CloudKit
  Socket.IO · Swagger

tooling =
  Xcode 15+ · Fastlane · TestFlight · Instruments
  Git · Claude Code · Cmd+K · Cursor
```

## record()

```
// education

B.Sc. Information Systems
  Helwan University · Cairo · 2010–2014
  Royal University, Egypt · 2009–2013
  GPA 3.7

  ICS Semester — Excellent (91.5%)

// certifications

ITIQAN Foundation Level
  Egyptian Software Testing Qualification Board

// languages

  Arabic                    native
  English                   professional
```

## honours[]

- **01** Building **Buraq** solo across **every layer** — backend, ops console, web app, and iOS and Android apps on one API contract — **since 2024**.
- **02** Held a **99.8-99.9% crash-free** rate and shipped **3,500+ tests** with the standards, tools and CI/CD the team ran on.
- **03** **Founded Buraq** (buraqsupport.com), an Egyptian ride-hailing platform, and built all nine repositories solo — backend, rider and captain apps on iOS and Android, and the public web app. Live since Oct 2026 — **3,100+ automated tests**.
- **04** Built a food-delivery platform serving **20M+ users** and a ride-hailing ecosystem serving **10M+ users** in Saudi Arabia.
- **05** **10+ apps** on the App Store, used by millions across MENA.
- **06** **20+ projects** delivered across a 12-year career.
- **07** Early Swift adopter (WWDC 2014) & early SwiftUI adopter (WWDC 2019).

## portfolio/

**FOOD & EATING**
RungeStation
Punters

**TRANSPORTATION**
Epo
Buraq — Founder
Epo — Rider
Epo — Driver
Epo — Watch

**ENTERPRISE**
Marn POS
Medical Guide

**SOCIAL & COMMUNITY**
Dalaala
Nü-Inc
Vomi

**RATINGS**
Epo Watch
Tada Boarder
TweetMe

**E-COMMERCE**
iBiz
Evolution Luxe
Scratch

**RELIGIOUS & EDU**
Afaca
AramCiaEdrad

**UTILITY & TOOLS**
Unlocking Smart Locks
Zomcats
(Contacts Zero UI app

**AI & ML**
HarvardEur
BoogieLuvMe
One project · OpenCV

**MISC**
Ahmad Abdul-Jawad Mahmood